

Basic Principles of Medicine 1

Module: Cardiovascular, Pulmonary, Renal

Lecture: CPR 49

Renal Clearance

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Recommended Reading

Medical Physiology: Principles for Clinical Medicine
with Companion Website Access, R. Rhoades & D. Bell
6th ed., 2023; Chapter 22

Principles of Pharmacology: The Pathophysiologic Basis of
Drug Therapy, D. Golan, E. J. Armstrong, & A. W.
Armstrong, 4th ed., 2016; p. 474

Learning Objectives

SOM.MK.II.BPM1.3.CPR.2.PHYS.0330. Explain the clearance principle. Use the clearance equation and identify appropriate compounds used to estimate the glomerular filtration rate and renal plasma flow.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0331. Distinguish between the use of inulin and creatinine clearances as measures of the glomerular filtration rate.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0332. Given the plasma and urine concentrations and the urine flow rate, calculate the filtered load, tubular transport, excretion rate, and clearance of inulin, creatinine, para-amino hippuric acid (PAH), glucose. Predict how changes in filtration, reabsorption, and secretion will affect renal excretion of each compound.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0333. Discuss the renal handling of glucose under normal conditions and in patients with diabetes mellitus.

Learning Objectives

SOM.MK.II.BPM1.3.CPR.2.PHYS.0334. Interpret clearance graphs for para-amino hippuric acid (PAH) and glucose at various plasma concentrations of these substances

SOM.MK.II.BPM1.3.CPR.2.PHYS.0381. With the help of a graph, explain tubular load, tubular transport maximum (T_{max}), renal threshold and splay for glucose.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0335. Discuss the renal handling of PAH under normal conditions and its role in renal plasma flow.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0336. Calculate osmolar and free water clearance and explain its significance. Explain expected free water clearance for an individual producing either a dilute or concentrated urine

Renal Clearance

With one of the kidneys' core functions being their ability to remove substances from plasma, renal clearance is defined as the volume of plasma cleared of a substance by the kidneys per unit time.

Renal clearance can be used to ascertain GFR and renal plasma flow (RPF), as well as renal handling of various substances.

Clearance is dependent on the plasma concentration of the substance, as well as the urine concentration of the substance and the urine flow rate. It is expressed as units of volume per minute (e.g. **mL/min**).

For substance 'x', the clearance is given as:

$$C_x = \frac{U_x \times V}{P_x}$$

Where,

C_x is the clearance of x

U_x is the urine concentration of x

V is the urine flow rate

Excretion rate and clearance are not the same. Excretion rate = $U_x \times V$

Calculating Clearance

Example: If the plasma concentration of substance 'x' is 0.2 mg/mL, the urine concentration is 6 mg/mL, and the urine flow rate is 2 mL/min,

(1) What is the excretion rate of x?

(2) What is the clearance of x?

$$\begin{aligned} \text{(1) Excretion rate} &= U_x \times V \\ &= 6 \text{ mg/mL} \times 2 \text{ mL/min} = \mathbf{12 \text{ mg/min}} \end{aligned}$$

$$\begin{aligned} \text{(2) Clearance of x (C}_x\text{)} &= \frac{U_x \times V}{P_x} \\ &= \frac{12 \text{ mg/min}}{0.2 \text{ mg/mL}} = \mathbf{60 \text{ mL/min}} \end{aligned}$$

How do filtration, reabsorption, secretion, and excretion influence clearance?

If a substance is filtered but is not excreted in urine, its clearance is zero (e.g., glucose *).

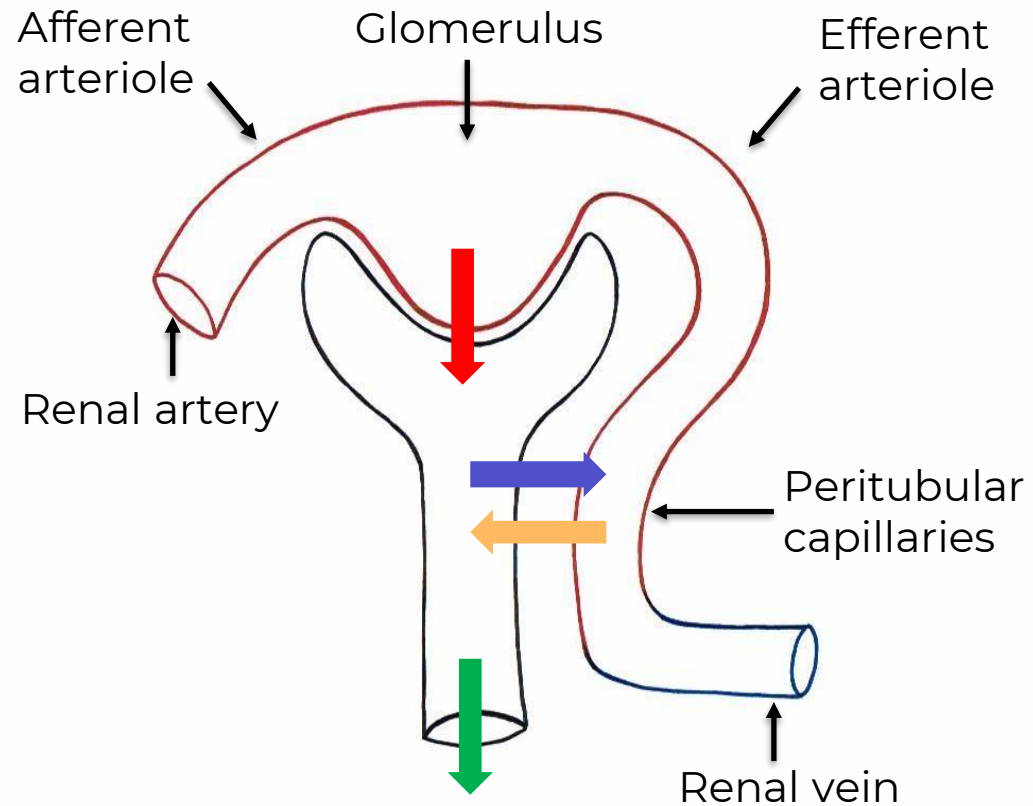
If a substance is filtered but is not reabsorbed from or secreted into the tubules, its clearance is equal to the GFR (e.g., inulin).

If the filtered load of a substance is greater than its excretion (i.e., net reabsorption), its clearance is less than that of inulin (e.g., sodium).

If the filtered load of a substance is less than its excretion (i.e., net secretion), its clearance is greater than that of inulin (e.g., para-amino hippurate – PAH).

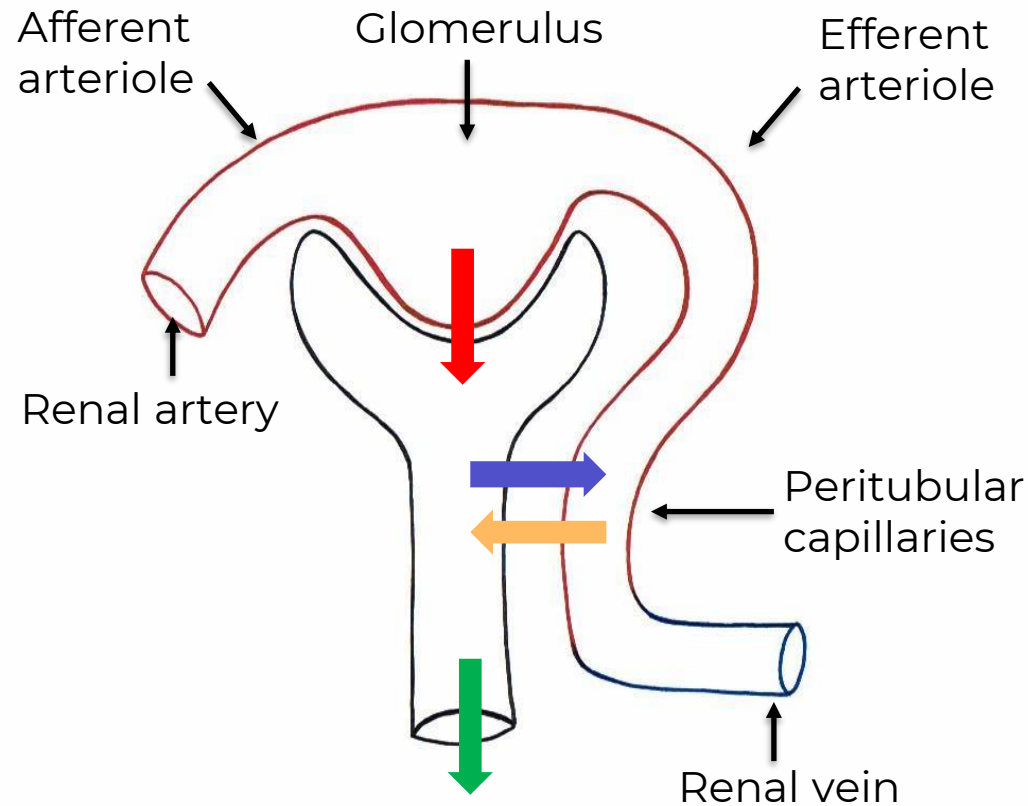
**Glucose can appear in urine in certain conditions; in such conditions, glucose clearance is NOT zero (e.g., diabetes mellitus, pregnancy).*

Representative Nephron



- Filtration
- Reabsorption
- Secretion
- Excretion

Quantifying Tubular Function



Example for substance 'z':

$$\text{Filtered load} = \text{GFR} \times P_z$$

$$\text{Reabsorption} = (\text{GFR} \times P_z) - (U_z \times V)$$

$$\text{Secretion} = (U_z \times V) - (\text{GFR} \times P_z)$$

$$\text{Excretion} = U_z \times V$$

$$\text{Excretion} = \text{Filtration} - \text{reabsorption} + \text{secretion}$$

Using Clearance to Measure GFR

GFR is an index of renal function.

Using clearance to measure GFR necessitates using a substance that possesses the following features:

- Freely filtered
- Not reabsorbed or secreted
- Not synthesized by the kidneys
- Non-toxic

It, therefore, implies that the amount of this substance excreted per minute equals the amount filtered per minute. Using 'a' as the substance:

Urine conc. of 'a' x urine flow rate = glomerular filtrate conc of 'a' x GFR

$$U_a \times V = GF_a \times GFR$$

$$GFR = \frac{U_a \times V}{GF_a} = \frac{U_a \times V}{P_a}$$

$$GF_a = P_a$$

'a' is a freely filtered substance with equal concentration in plasma and the filtrate.

Inulin Clearance as a Measure of GFR

Inulin, a fructose polymer, is freely filtered, not reabsorbed or secreted, not synthesized by the kidneys, and is non-toxic.

Inulin is cleared by filtration alone, making it the ideal substance for measuring GFR. It is the gold standard for GFR measurement.

Therefore, **inulin clearance = GFR**

$$\text{GFR} = \frac{U_{\text{In}} \times V}{P_{\text{In}}}$$

Creatinine Clearance as a Measure of GFR

Creatinine, an end product of muscle metabolism, is an alternative used clinically to measure GFR. It is produced at a relatively constant rate and is excreted in urine.

Potential drawbacks of creatinine clearance for GFR measurement:

- It is secreted by the kidneys resulting in increased urine excretion. This is capable of increasing the numerator of the clearance formula by up to 20%.
- The colorimetric method used for plasma creatinine measurement measures other substances like glucose, resulting in about a 20% increase in the denominator of the clearance formula.

With a 20% increase in both the numerator and denominator of the formula, both drawbacks cancel out each other, making the clearance of creatinine a reasonable estimation of GFR.

$$C_{cr} = \frac{U_{cr} \times V}{P_{cr}}$$

Creatinine clearance may overestimate GFR in a patient with chronic kidney disease with a low GFR.

Calculating Creatinine Clearance

Example: A patient evaluated for renal disease has the following laboratory values:

- 24-hour urine volume = 600 mL
- Plasma creatinine = 10 mg/dL
- Urine creatinine = 240 mg/dL
- BUN = 80 mg/dL (normal 10-20 mg/dL)

Calculate creatinine clearance

$$C_{cr} = \frac{240 \text{ mg/dL} \times 600 \text{ mL/day} (0.42 \text{ mL/min}^*)}{10 \text{ mg/dL}}$$

$$= 10 \text{ mL/min} \quad (\text{normal } 90\text{-}140 \text{ mL/min})$$

$$C_{cr} = \frac{U_{cr} \times V}{P_{cr}}$$

** 0.42 mL/min is derived from converting 600 mL/day to mL/min*

$$\frac{600}{24 \times 60} = 0.42$$

To convert mL/day to mL/min, divide by 24 (hours in the day) and 60 (minutes in an hour)

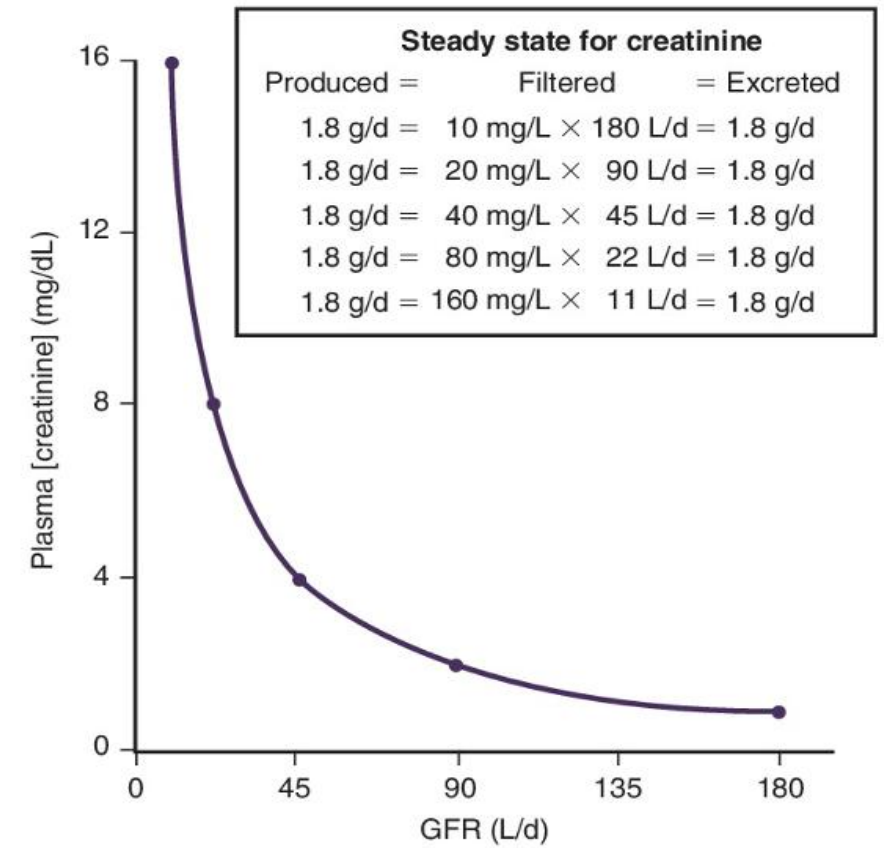
A Low GFR is suggestive of renal insufficiency

Inverse Relationship of Plasma Creatinine Concentration and GFR

When the production and excretion of creatinine are equal (steady state), a **decrease in GFR results in an increase in plasma creatinine concentration.**

Considerations:

- A change in GFR does not result in an immediate detectable change in plasma creatinine concentration. It may take days before changes are detected.
- Increased muscle mass increases plasma creatinine concentration. There are also age and sex differences in plasma creatinine concentration due to the muscle mass differences.
- Drugs that inhibit tubular secretion of creatinine cause an increase in plasma creatinine.



Source: Medical Physiology: Principles for Clinical Medicine, 6e, 2023. Copyright © Wolters Kluwer

Using Clearance to Measure RPF and RBF

RBF (renal blood flow) is defined as the volume of blood delivered to the kidneys per unit time (approx. 1200 mL/min). RPF (renal plasma flow) is the volume of plasma that flows through the kidneys per unit time.

The clearance of para-amino hippurate (PAH) = RPF

$$\text{RBF} = \frac{\text{RPF}}{1 - \text{hematocrit}}$$

When PAH, an organic anion, is administered intravenously, it is filtered at the glomerular capillaries and rapidly secreted into the tubular fluid, resulting in an almost total clearance from the plasma flowing through the kidneys (RPF).

Calculating PAH Clearance and RPF

The amount of PAH entering the kidneys through plasma per minute equals the amount of PAH excreted by the kidneys in urine per minute.

$$\text{RPF} \times P_{\text{PAH}} = U_{\text{PAH}} \times V$$

$$\text{RPF} = \frac{U_{\text{PAH}} \times V}{P_{\text{PAH}}}$$

$$\text{RPF} = C_{\text{PAH}}$$

Given:

Urine flow of 1 mL/min, U_{PAH} of 65 mg/mL, and P_{PAH} of 0.1 mg/mL, calculate the renal plasma flow.

$$\text{RPF} = \frac{U_{\text{PAH}} \times V}{P_{\text{PAH}}} = C_{\text{PAH}}$$

$$\begin{aligned} \text{RPF} &= \frac{65 \text{ mg/mL} \times 1 \text{ mL/min}}{0.1 \text{ mg/mL}} \\ &= 650 \text{ mL/min} \end{aligned}$$

Plasma PAH Concentration and PAH Clearance

At low plasma concentrations, all PAH is cleared from plasma, and renal venous PAH is zero.

Therefore, the volume of plasma cleared of PAH = volume of plasma entering or leaving the kidneys = RPF.

$$\mathbf{RPF = C_{PAH}}$$

If the P_{PAH} increases, the filtered load increases. The PAH secretion into tubular fluid also increases but is limited when the **maximum secretion rate** is reached. Under these circumstances, not all PAH is removed in the kidney, resulting in a **progressive fall in C_{PAH}**

Fractional Excretion

The proportion of the filtered load of a substance excreted in urine is the fractional excretion of that substance.

For substance 'x':

$$FE_x = \frac{U_x \times V}{GFR \times P_x}$$

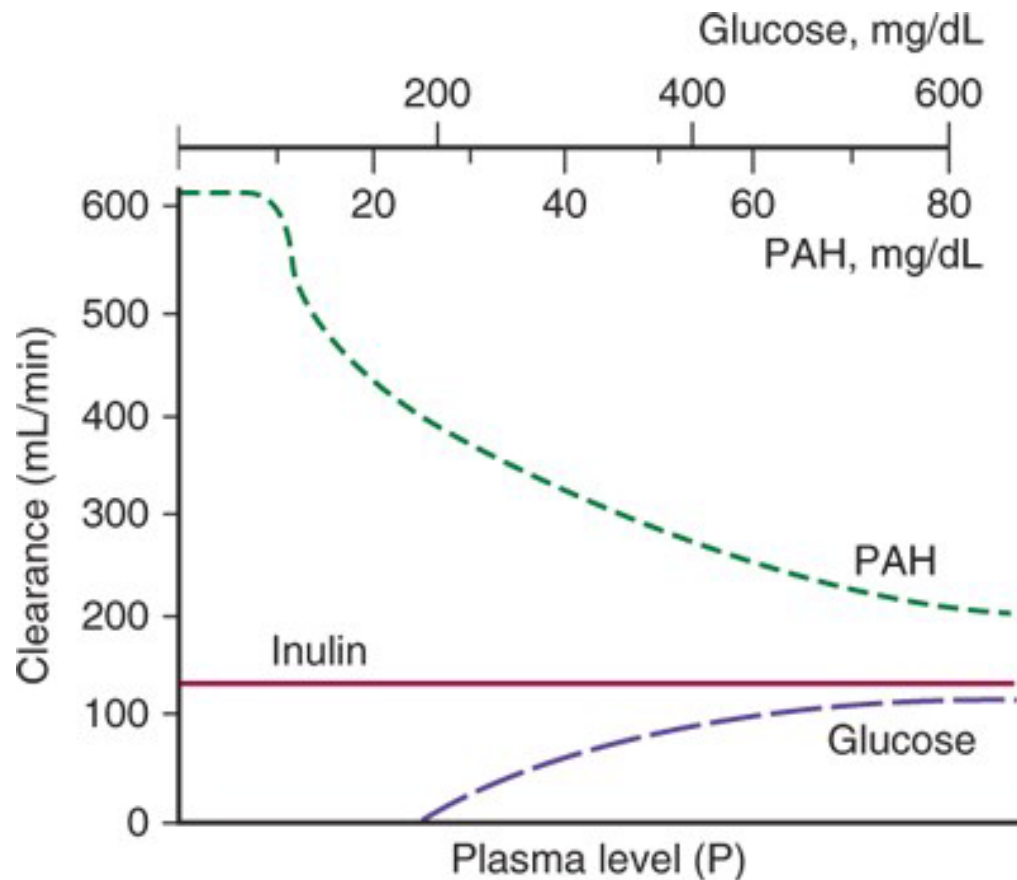
Fractional Excretion of Sodium (FE_{Na})

Can be used to ascertain if the cause of acute kidney injury (AKI) is due to prerenal or renal pathology.

FE_{Na} less than 1% is indicative of prerenal pathology. This indicates that the kidney's function as sodium reabsorption is normal.

FE_{Na} greater than 2% suggests renal pathology. This indicates that the kidney's ability to reabsorb sodium is reduced by tubular dysfunction due to kidney injury.

Plasma Concentration-Clearance Relationship (Inulin, Glucose, PAH)



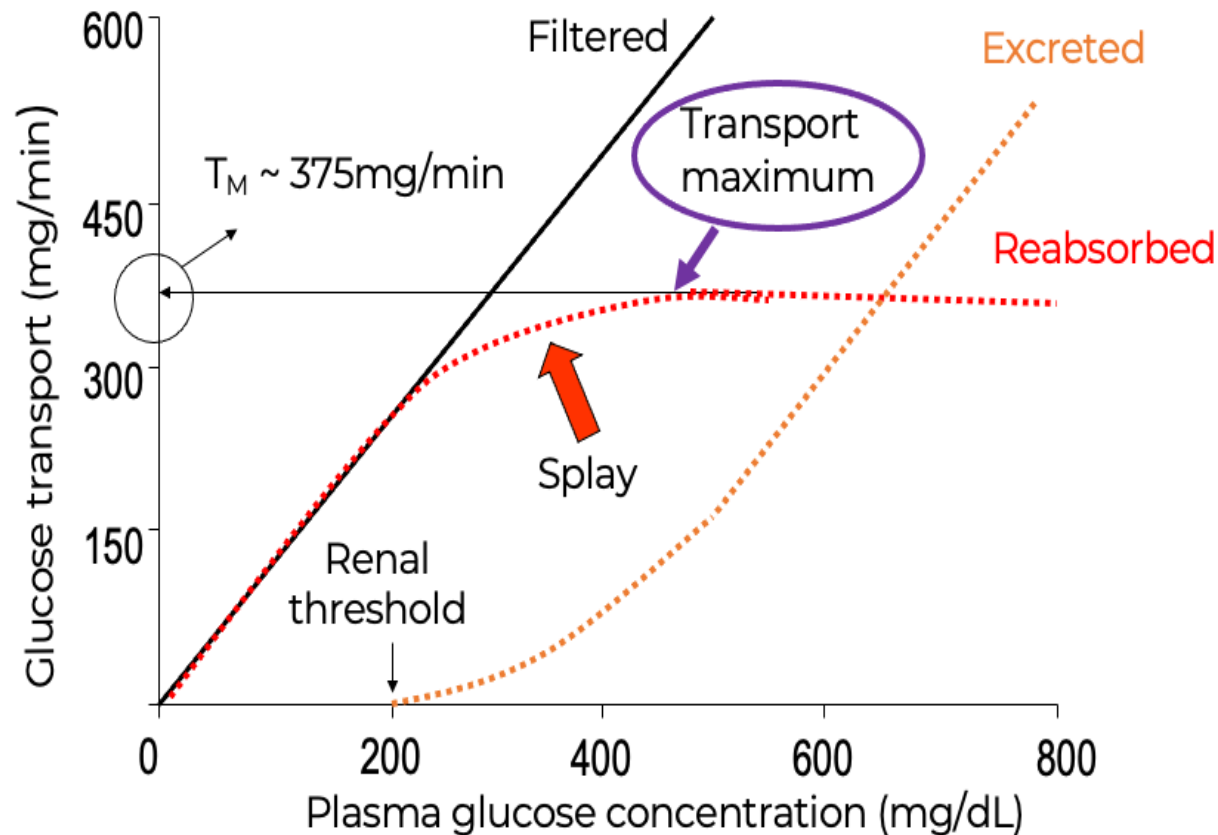
Source: K.E. Barrett, S.M. Barman, H.L. Brooks, Jason X.J. Yuan:
Ganong's Review of Medical Physiology, Twenty-Sixth Edition
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Inulin – Clearance remains the same regardless of plasma concentration.

Glucose – Clearance is zero when plasma concentration is below the renal threshold but starts to increase when the plasma concentration is above the renal threshold.

PAH – Clearance is high when the plasma concentration is low but decreases progressively as the plasma concentration exceeds the maximum secretion rate.

Renal Handling of Glucose



At sub-threshold plasma concentrations, all of the filtered glucose is reabsorbed, and none is excreted in urine.

Glucose is excreted in urine when the renal threshold is reached.

The renal threshold for glucose is the plasma glucose concentration at which glucose starts getting excreted in urine.

Splay implies that reabsorption does not suddenly reach the transport maximum. The reason for this is that nephrons have different transportation abilities – nephron heterogeneity

Quantifying the Kidney's Concentration Ability

Free water clearance (C_{H_2O}) is the volume of solute-free water saved or excreted by the kidneys per unit time.

$$C_{H_2O} = V - C_{osm}$$

where

$$\left[C_{osm} = \frac{U_{osm} \times V}{P_{osm}} \right]$$

Note:

V = Urine flow rate (mL/min)

C_{osm} = Osmolal clearance

U_{osm} = Urine osmolality

P_{osm} = Plasma osmolality

Implications of Free Water Clearance Values

If C_{H_2O} is 0 (zero), urine is isosmotic to plasma.

If C_{H_2O} is a positive value, the kidneys are excreting excess water (urine is dilute, and ADH is low).

If C_{H_2O} is a negative value, the kidneys are conserving water (urine is concentrated, and ADH is high).

What is the implication of C_{H_2O} of 1?

What is the implication of C_{H_2O} of -1?

References

- Medical Physiology: Principles for Clinical Medicine
R. Rhoades & D. Bell, 6th ed.
- Ganong's Review of Medical Physiology, 26th ed.